

Extraction of Critical Metals from Spent LiFePO_4 (LFP) Batteries using Polyol-based Metallurgical Method

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Introduction

- Extraction of Lithium and critical metals from spent Li-ion batteries is essential for resource sustainability and environmental protection
- End-of-life batteries (Dead and spent Li-ion) are dismantled and processed into black mass, then pyrolyzed at 500 °C in N_2 atm
- Conventionally, hydrometallurgical and pyrometallurgical methods are used to extract Li and critical metals, while relying on strong or toxic acids and high temperatures.
- Here, we demonstrate a green polyol-based metallurgical approach using a Critic Acid: Ethylene Glycol (CAEG) system for Lithium extraction from spent battery LiFePO_4 Black Mass (LFP-BM)

Objective

- Recovery of valuable metals, reduce environmental impact, and minimize waste generation
- Enhance Selectivity: Selectively precipitate Fe and P while keeping Li in the solution for easy separation
- Optimize conditions for maximum recovery

Methodology

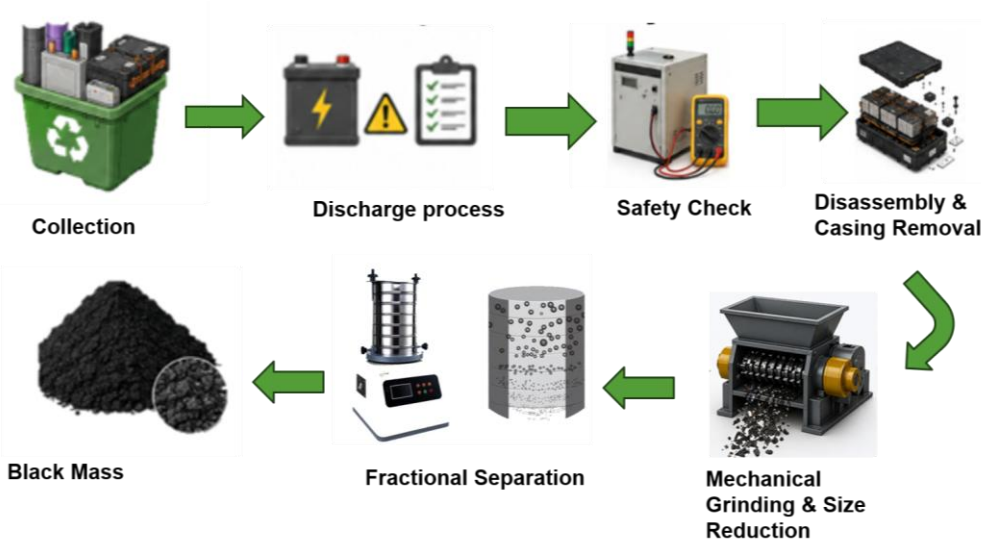


Figure 1. Schematic representation of generation of Black Mass from Spent Li ion battery

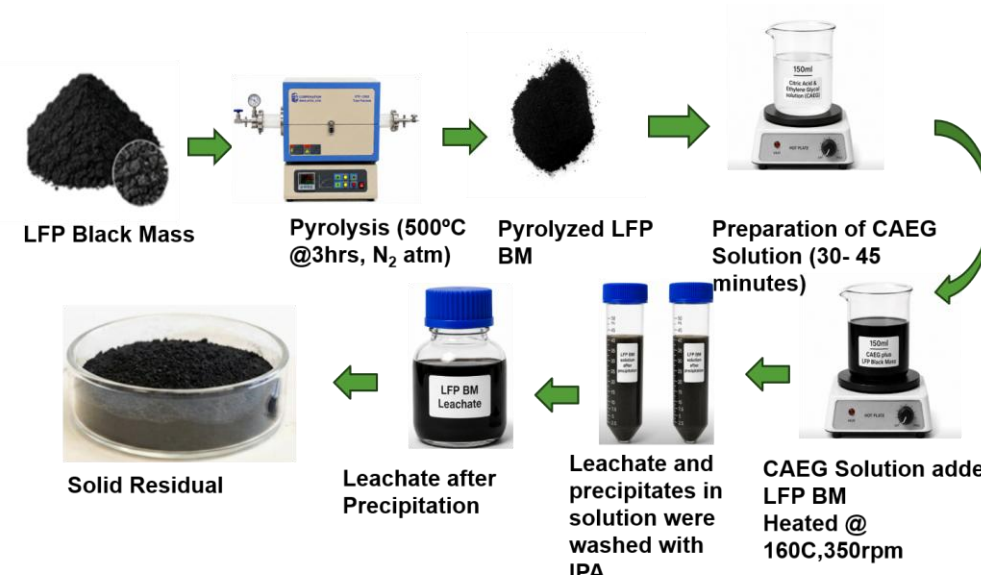
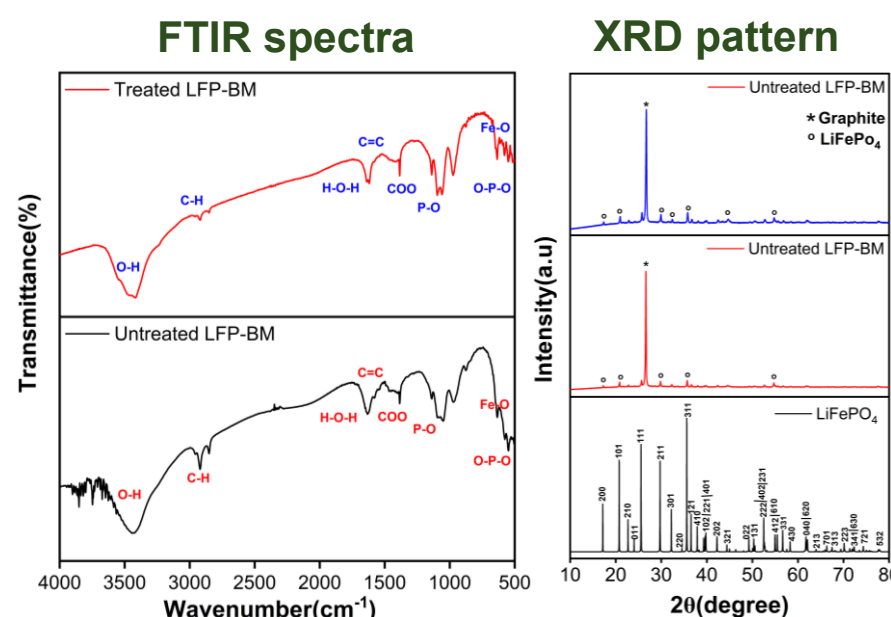
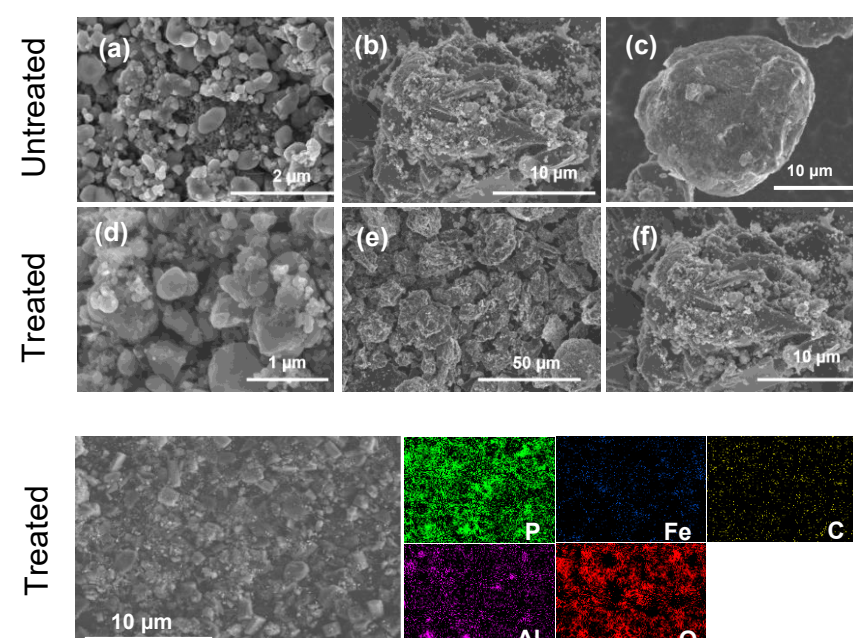


Figure 2. Schematic representation of Polyol-based metallurgical process

Characterization of LFP Black Mass

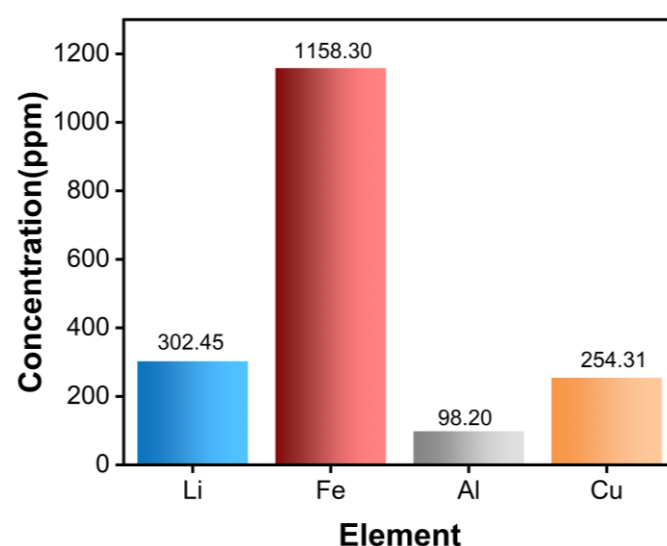


Electron Micrographs & Elemental Mapping

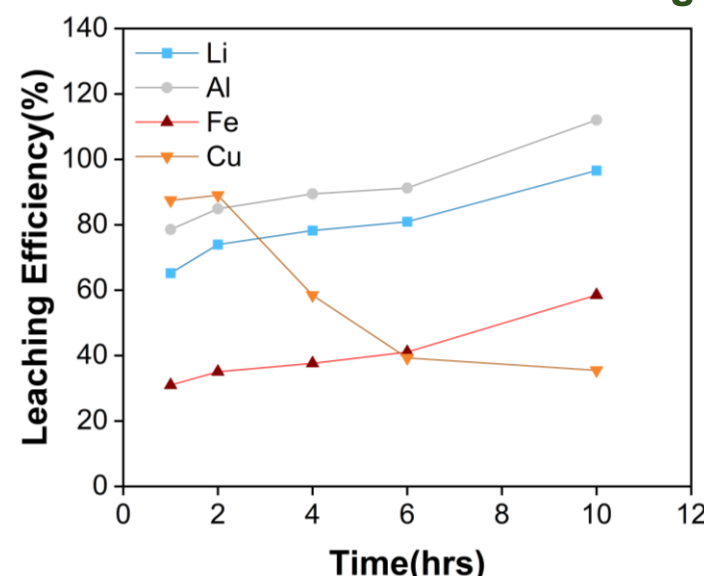


Leaching LFP Black Mass

ICP of LFP Black mass

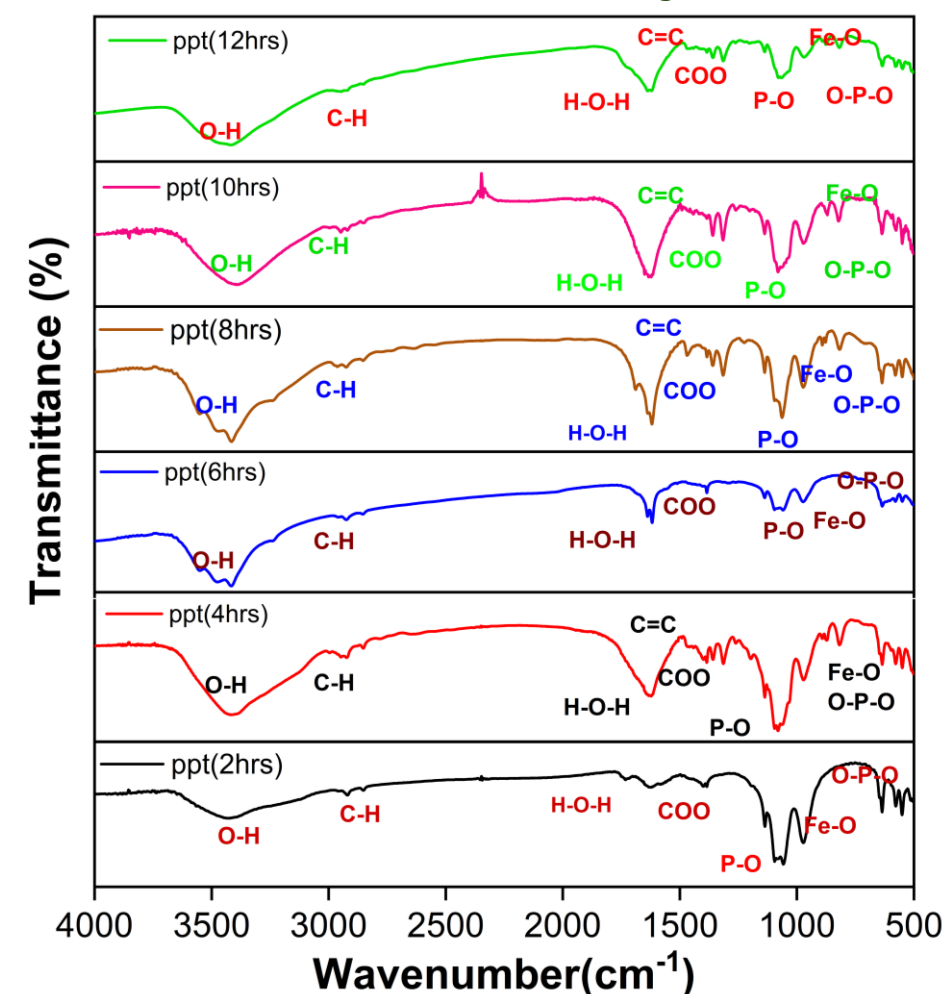


Effect of duration of Leaching

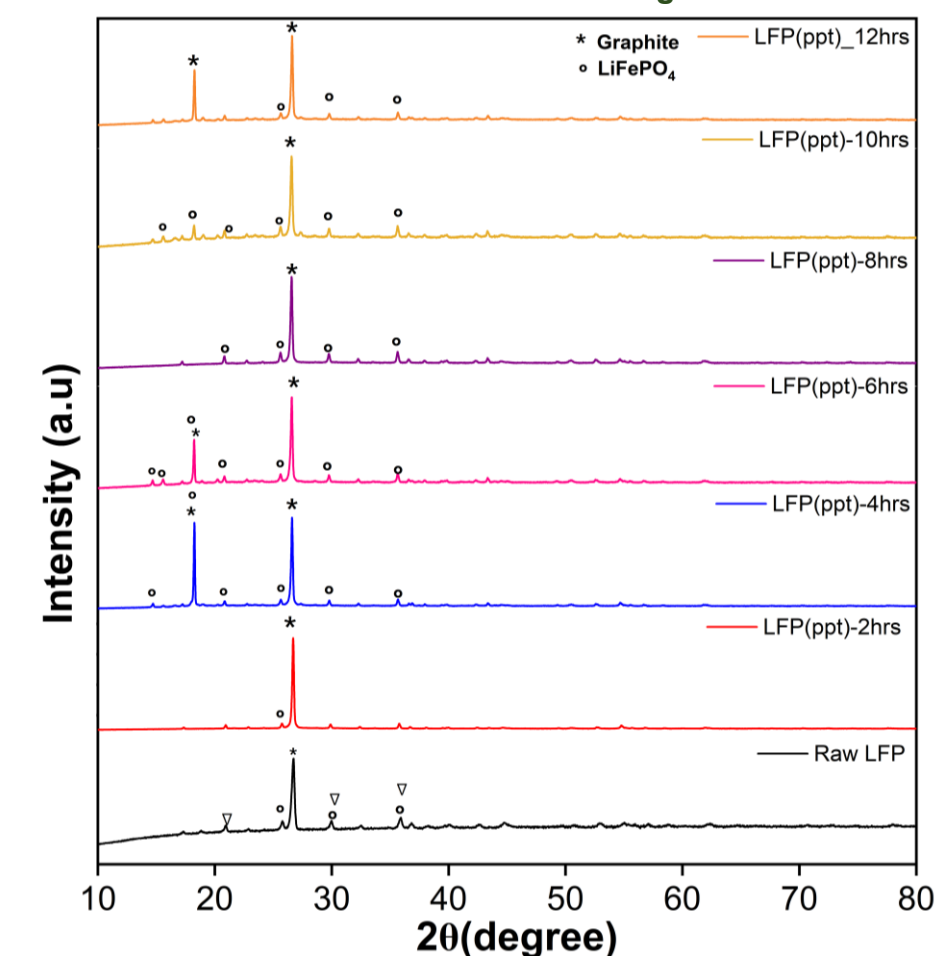


Results and Discussion

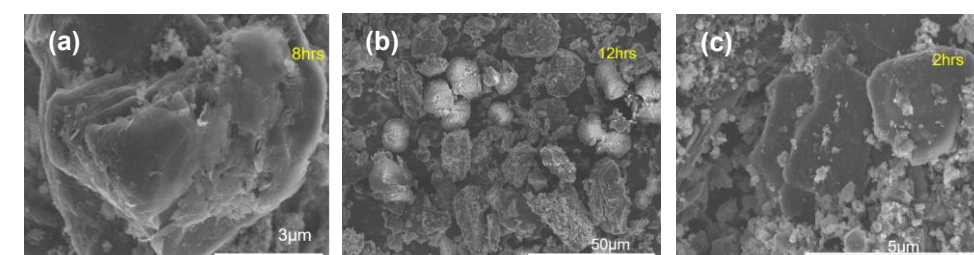
FTIR Spectra of Residue Collected at Different Durations of Leaching



XRD Pattern of Residue Collected at Different Durations of Leaching



Morphology of Residue collected at different durations



Conclusion

- LFP black mass characterization (XRD & ICP-MS) confirms the presence of LiFePO_4 and associated metal composition
- Efficient Li recovery (>90%) is achieved using the polyol-based green leaching method
- Selectivity studies of the CAEG system are required to understand Li extraction over Fe, P, and other elements

Reference: Yu, L., Bai, Y., Essehli, R., Bisht, A., & Belharouak, I. (2023). Efficient separation and coprecipitation for simplified cathode recycling. *Energy Storage Materials*

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